

YAKIMA, WA, USA

WMO: 727810

Lat: 46.568N

Lon: 120.543W

Elev: 324

StdP: 97.49

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24243

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.7	-9.8	-17.0	0.9	-9.7	-14.1	1.2	-8.0	10.4	7.1	8.4	7.7	2.0	270	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	36.3	18.9	34.5	18.5	32.6	17.7	20.2	33.2	19.2	32.1	18.4	30.9	2.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.4	24.4	14.2	10.5	23.5	13.1	9.8	23.4	59.0	33.4	55.7	32.2	53.0	30.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.1	DB	-17.0	39.1	4.5	1.5	-20.2	40.1	-22.8	41.0	-25.4	41.9	-28.6	42.9	(4)
(5)			WB	-17.2	22.6	4.1	2.7	-20.1	24.6	-22.6	26.2	-24.9	27.7	-27.9	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	0.0	2.5	6.4	10.0	14.9	18.5	22.8	21.7	16.8	9.9	3.6	-0.8		
	DBStd	8.90	4.22	4.07	3.51	3.55	4.13	3.91	3.60	3.37	3.64	3.78	4.35	4.22		
	HDD10.0	1269	311	211	118	42	6	0	0	0	2	47	196	336		
	HDD18.3	3183	569	443	369	249	122	44	6	9	72	263	442	594		
	CDD10.0	1474	0	1	7	44	159	254	397	361	204	43	4	0		
	CDD18.3	347	0	0	0	1	16	49	144	112	24	1	0	0		
	CDH23.3	5839	0	0	1	38	344	829	2264	1782	559	22	0	0		
	CDH26.7	2787	0	0	0	7	118	349	1209	891	208	4	0	0		
(14) Wind	WSAvg	2.6	1.8	2.3	2.9	3.2	3.1	3.3	3.0	2.9	2.7	2.4	2.0	1.7		
(15) Precipitation	PrecAvg	199	29	20	17	13	15	13	5	8	8	14	24	35		
	PrecMax	336	93	62	52	41	65	64	18	53	53	62	72	106		
	PrecMin	106	2	0	0	0	0	0	0	0	0	0	0	2		
	PrecStd	47	19	15	13	10	14	13	5	12	9	12	16	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	16.3	21.3	27.6	33.1	36.9	38.9	37.9	34.4	26.2	18.4	12.5		
		MCWB	7.9	9.7	11.7	14.5	16.7	19.2	19.8	19.6	18.5	14.9	12.8	8.2		
	2%	DB	10.2	13.2	18.3	23.7	30.0	33.2	36.9	35.7	31.3	22.4	15.1	9.6		
		MCWB	6.0	7.5	10.0	13.2	16.0	18.0	19.0	18.9	17.3	12.9	9.7	5.9		
	5%	DB	7.8	11.0	16.0	20.9	27.2	30.5	35.1	33.8	29.1	20.0	12.9	7.2		
		MCWB	4.7	6.0	8.8	11.5	14.9	16.5	18.6	18.3	16.3	11.8	8.2	4.3		
	10%	DB	5.6	9.2	14.1	18.4	24.5	28.2	33.2	31.8	26.6	17.9	10.9	5.1		
		MCWB	3.4	4.9	7.7	10.1	13.7	15.6	17.9	17.5	15.2	10.7	6.9	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	10.3	12.4	15.7	18.0	20.5	22.1	21.3	19.5	15.6	13.0	8.8		
		MCDB	12.2	15.8	19.6	25.1	29.9	33.8	34.7	33.7	31.3	23.8	17.6	12.1		
	2%	WB	6.3	7.9	10.8	13.8	16.7	18.5	20.4	19.9	18.2	13.9	10.4	6.4		
		MCDB	9.5	12.2	17.2	22.6	27.9	31.4	33.4	33.0	29.8	20.8	14.3	9.2		
	5%	WB	5.0	6.6	9.5	12.0	15.6	17.3	19.5	19.0	17.0	12.6	8.9	4.7		
		MCDB	7.6	10.3	15.3	19.7	25.5	29.3	32.7	31.6	27.3	18.7	12.2	6.8		
	10%	WB	3.6	5.4	8.2	10.6	14.4	16.2	18.6	18.1	15.9	11.4	7.4	3.1		
		MCDB	5.3	8.7	13.5	17.5	23.2	26.9	31.4	30.3	25.1	16.9	10.4	5.0		
(35) Mean Daily Temperature Range	MDBR		7.6	10.6	13.4	14.8	15.5	16.1	18.2	18.3	17.9	14.9	11.2	7.5		
	5% DB	MCDBR	11.4	13.6	16.3	18.5	19.6	19.3	20.5	20.7	21.1	17.4	13.2	11.1		
		MCWBR	8.1	8.6	9.4	9.5	8.2	7.5	7.3	7.9	9.2	9.5	8.7	8.0		
	5% WB	MCDBR	10.7	12.4	15.2	17.3	17.5	18.2	19.2	19.4	19.5	15.8	12.3	10.3		
		MCWBR	7.7	8.1	9.1	9.2	7.6	7.5	7.5	7.8	8.8	9.2	8.5	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.299	0.305	0.325	0.336	0.336	0.339	0.337	0.330	0.319	0.308	0.288		
	taud		2.485	2.514	2.541	2.473	2.466	2.486	2.488	2.489	2.496	2.521	2.496	2.494		
	Ebn at Noon		813	885	931	934	929	928	920	910	887	847	785	777		
	Edh at Noon		65	77	87	102	107	105	104	99	90	76	64	58		
(44) All-Sky Solar Radiation	RadAvg		1.21	2.19	3.36	4.92	5.99	6.84	7.52	6.35	4.67	2.77	1.47	0.97		
	RadStd		0.15	0.26	0.27	0.34	0.42	0.39	0.36	0.36	0.29	0.30	0.16	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.57	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	+155	+81
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+37

Nomenclature: See separate page